

EXACT VERIFICATION NOTES FOR 38-, 39-, 40-, 42-, AND 50-VERTEX COUNTEREXAMPLES TO WOW-284

SAMUIL PETKOV

ABSTRACT. This working note verifies an induced 40-vertex counterexample to the Graffiti conjecture WOW-284 and records smaller induced descendants. The 40-vertex $(6, 5)$ -cage has minimum dual degree 6 and least distance eigenvalue -5 . Deleting any edge endpoints produces a 38-vertex graph with

$$\delta^* = \frac{17}{3}, \quad \lambda_{\min}(D) = -3 - \sqrt{7}, \quad \Phi = \frac{8}{3} - \sqrt{7} > 0.$$

Exact exhaustive computation shows that all 120 edge-endpoint deletions have the same distance characteristic polynomial; all 40 single-vertex deletions are also strict counterexamples. We prove a spectral criterion for every regular girth-at-least-five graph of diameter three, derive a Moore second-subconstituent construction, and record both positive and negative generalization tests. No minimality, first-discovery, or unconditional infinite-family claim is made.

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1. DEFINITIONS AND SCOPE

For a finite simple connected graph G , write $d(v)$ for vertex degree and

$$d^*(v) = \frac{1}{d(v)} \sum_{u \in N(v)} d(u), \quad \delta^*(G) = \min_{v \in V(G)} d^*(v).$$

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Let $D(G)$ be the ordinary graph-distance matrix and let $\partial_n(G) = \lambda_{\min}(D(G))$. WOW-284 asserts that every connected graph of order at least three and girth at least five satisfies

$$\delta^*(G) \leq -\partial_n(G)$$

[1, Conjecture 7.16]. The counterexample score is

$$\Phi(G) = \delta^*(G) + \lambda_{\min}(D(G)).$$

A strict counterexample has $\Phi(G) > 0$.

The statements below have two independent forms of verification. The analytic calculations use exact matrix identities and characteristic polynomials. The accompanying programs reconstruct graph distances by integer breadth-first search and use exact rational or integer arithmetic. Floating-point computation is used only in the explicitly labelled three-vertex exploratory screen.

2. THE AMBIENT 50-VERTEX COORDINATE GRAPH

All coordinates lie in $\mathbb{F}_5$. Introduce vertices

$$\{P_{i,j} : i, j \in \mathbb{F}_5\} \cup \{Q_{k,\ell} : k, \ell \in \mathbb{F}_5\}$$

and edges

$$P_{i,j} \sim P_{i,j+1}, \quad Q_{k,\ell} \sim Q_{k,\ell+2}, \quad P_{i,j} \sim Q_{k,ik+j}.$$

This is the Hoffman–Singleton graph, although no named-graph theorem is needed for the finite certificates. Its adjacency matrix A_{HS} satisfies

$$A_{HS}^2 = 6I - A_{HS} + J. \tag{1}$$

Indeed, adjacent vertices have no common neighbor and nonadjacent vertices have exactly one. Thus it is 7-regular, has girth five and diameter two, and

$$\text{Spec}(A_{HS}) = \{7^{(1)}, 2^{(28)}, (-3)^{(21)}\}.$$

Since $D = 2J - 2I - A_{HS}$,

$$\text{Spec}(D) = \{91^{(1)}, 1^{(21)}, (-4)^{(28)}\}.$$

Hence $\delta^* = 7$ and $\Phi = 3$. The classical construction and Moore graph context originate with Hoffman and Singleton [2]; the distance spectrum was previously recorded in the cage literature [3].

3. THE 40-VERTEX REGULAR COUNTEREXAMPLE

Delete the ten vertices

$$S = \{P_{0,j}, Q_{0,j} : j \in \mathbb{F}_5\}$$

and call the remaining induced graph R . The deleted vertices induce a Petersen graph: the P -vertices and Q -vertices form two 5-cycles and the cross edges $P_{0,j}Q_{0,j}$ form a perfect matching. This realizes the classical 40-vertex $(6, 5)$ -cage [4, 5, 6, 7].

Theorem 3.1. *The graph R is simple, connected and 6-regular on 40 vertices, has girth five and diameter three, and has spectra*

$$\text{Spec}(A(R)) = \{6^{(1)}, 2^{(18)}, 1^{(4)}, (-2)^{(5)}, (-3)^{(12)}\},$$

$$\text{Spec}(D(R)) = \{75^{(1)}, 3^{(5)}, 0^{(16)}, (-5)^{(18)}\}.$$

Consequently

$$\delta^*(R) = 6, \quad \lambda_{\min}(D(R)) = -5, \quad \Phi(R) = 1.$$

Proof. Every surviving vertex loses exactly one neighbor in S , so R is 6-regular and has $40 \cdot 6/2 = 120$ edges. It is induced in a girth-five graph and retains a full P -layer 5-cycle, so its girth is exactly five.

Order the ambient adjacency matrix as

$$A_{HS} = \begin{pmatrix} B & C \\ C^\top & P \end{pmatrix},$$

where $B = A(R)$ and P is the Petersen adjacency matrix. Each deleted vertex has four outside neighbors, while two distinct deleted vertices have no common outside neighbor. Hence

$$C^\top C = 4I_{10}.$$

Taking the upper-right block of (1) gives

$$BC = C(J - I - P). \quad (2)$$

The columns of C are independent. Since

$$\text{Spec}(P) = \{3^{(1)}, 1^{(5)}, (-2)^{(4)}\},$$

Equation (2) gives the spectrum $\{6^{(1)}, (-2)^{(5)}, 1^{(4)}\}$ on $\text{col}(C)$.

If $x \in \ker(C^\top)$, then $x \perp \mathbf{1}$, because $C\mathbf{1} = \mathbf{1}$. The upper-left block of (1) therefore gives

$$(B^2 + B - 6I)x = 0.$$

The remaining 30 eigenvalues are 2 or -3 . Dimension and trace yield multiplicities 18 and 12. In particular, the regular eigenvalue 6 is simple, so R is connected.

For completeness, the same block equations give a structural diameter bound. If two nonadjacent outside vertices have their unique ambient common neighbor outside S , their distance in R is two. If their unique common neighbor is the same vertex of S , then the corresponding entry of B^2 is zero and, using (2), the corresponding entry of B^3 is six. Thus every pair is at distance at most three. The Moore bound excludes diameter two, since $40 > 1 + 6 + 6 \cdot 5 = 37$. Hence the diameter is three.

For every connected k -regular graph of girth at least five and diameter three,

$$D = 3J + (k - 3)I - 2A - A^2; \quad (3)$$

this is proved in Section 7. Here $k = 6$. The principal distance eigenvalue is 75, while a nonprincipal adjacency eigenvalue θ maps to

$$3 - 2\theta - \theta^2 = 4 - (\theta + 1)^2.$$

Substitution gives the displayed distance spectrum. Regularity gives $\delta^*(R) = 6$, completing the proof. $\square$

4. THE 38-VERTEX EDGE-DELETION COUNTEREXAMPLE

Delete the adjacent vertices

$$a = P_{1,0}, \quad b = P_{1,1}$$

from R , and call the resulting graph H . Equivalently, delete from the 50-vertex coordinate graph

$$\{P_{0,j}, Q_{0,j} : j \in \mathbb{F}_5\} \cup \{P_{1,0}, P_{1,1}\}.$$

Relabeling the surviving ambient labels increasingly by $0, \dots, 37$, an independent graph6 specification is

```
egCGGc??G?_@?H????G?@??C?@HCP?AG_OAIaG@qGaBH?Q?HGA??c@@?HG?s_H?E_SA?
@OGC?@OI??OGD??WA_O?E@@@??CCE???GKC??AKCC??EAAA??B?
```

Theorem 4.1. *The graph H is simple and connected, has 38 vertices, 109 edges, girth five and diameter three, and has degree multiset $6^{(28)}, 5^{(10)}$. Moreover,*

$$\delta^*(H) = \frac{17}{3}, \quad \lambda_{\min}(D(H)) = -3 - \sqrt{7},$$

and therefore

$$\Phi(H) = \frac{8}{3} - \sqrt{7} > 0.$$

Proof. Induced deletion preserves simplicity and cannot create a triangle or 4-cycle. A distinct P -layer remains a 5-cycle. Exact breadth-first search from every vertex gives connectedness and diameter three.

Let

$$X = N_R(a) \setminus \{b\}, \quad Y = N_R(b) \setminus \{a\}.$$

The girth condition implies that X and Y are disjoint and that no vertex of $X \cup Y$ has a remaining neighbor in $X \cup Y$. Thus these ten vertices have degree five and all their remaining neighbors have degree six. Every other vertex has degree six. A degree-six vertex has at most one neighbor in X and at most one in Y , or a 4-cycle would result. If it has t such neighbors, where $t \in \{0, 1, 2\}$, then

$$d^*(v) = \frac{5t + 6(6 - t)}{6} = 6 - \frac{t}{6}.$$

There are $10 \cdot 5 = 50$ incidences from $X \cup Y$ to the 28 degree-six vertices, so some degree-six vertex has $t = 2$. Hence $\delta^*(H) = 17/3$.

The exact distance characteristic polynomial is recorded in Appendix A. It contains the factor $(x^2 + 6x + 2)^2$, whose roots are $-3 \pm \sqrt{7}$. An exact Sturm count shows that the complete characteristic polynomial has exactly one distinct root below $-28/5$. Since

$$-3 - \sqrt{7} < -\frac{28}{5} \iff 7 > \left(\frac{13}{5}\right)^2,$$

this is the least distance eigenvalue. Independently, an exact rational LDL^T decomposition of $3D(H) + 17I$ has 38 positive pivots; the smallest in the fixed graph6 order is

$$\frac{12434320241260060}{51172748480408387} > 0.$$

Finally,

$$\frac{8}{3} > \sqrt{7} \quad \text{because} \quad \frac{64}{9} > 7,$$

so the violation is strict. □

5. ORDER 39 AND ORDER 42

5.1. Deleting one vertex from the 40-vertex graph. Delete $P_{1,0}$ from R . Its six former neighbors have degree five and the other 33 vertices have degree six. A degree-six vertex is adjacent to at most one degree-five vertex, by the absence of 4-cycles. Since incidences exist, the minimum dual degree is

$$\delta^* = \frac{5 + 5 \cdot 6}{6} = \frac{35}{6}.$$

Exact rational factorization gives

$$6D + 35I \succ 0;$$

the smallest pivot in the fixed coordinate order is

$$\frac{128650413252103}{36129805596929} > 0.$$

Thus the 39-vertex graph is a strict counterexample. Its exact distance characteristic polynomial appears in Appendix A.

5.2. The second subconstituent of one ambient vertex. Fix $P_{0,0}$ in the Hoffman–Singleton graph and delete it together with all seven of its neighbors. The remaining 42 vertices form a 6-regular graph with

$$\begin{aligned}\text{Spec}(A) &= \{6^{(1)}, 2^{(21)}, (-1)^{(6)}, (-3)^{(14)}\}, \\ \text{Spec}(D) &= \{81^{(1)}, 4^{(6)}, 0^{(14)}, (-5)^{(21)}\}.\end{aligned}$$

Hence $\delta^* = 6$ and $\Phi = 1$. Section 8 places this example in a parameterized Moore-graph construction.

6. EXACT FINITE DESCENDANT FAMILIES

The scripts do more than verify one selected vertex and one selected edge. They exhaust the corresponding labelled deletion classes inside the fixed 40-vertex graph.

Proposition 6.1. *For every vertex v of R , the graph $R - v$ has $\delta^* = 35/6$ and the same exact distance characteristic polynomial P_{39} displayed in Appendix A. The polynomial has no root at or below $-35/6$, so all 40 labelled single-vertex deletions are strict counterexamples.*

For every edge uv of R , the graph $R - \{u, v\}$ has $\delta^ = 17/3$ and the same exact distance characteristic polynomial P_{38} . Its least root is $-3 - \sqrt{7}$, so all 120 labelled edge-endpoint deletions are strict counterexamples.*

Proof. The program `scripts/verify_descendant_families.py` enumerates all 40 vertices and all 120 edges. For each induced graph it reconstructs the integer distance matrix by breadth-first search, computes every dual degree as a rational number, and computes the characteristic polynomial over $\mathbb{Z}$. Equality with P_{39} or P_{38} is exact. One Sturm count for each common polynomial then proves the strict spectral bound for every member of the finite class. No graph-isomorphism assumption is used. $\square$

Remark 6.2. A numerical screen of all $\binom{40}{3} = 9880$ three-vertex deletions found no positive score; its best value was approximately -0.306979936667 . That screen is deliberately not promoted to an exact theorem and is not a minimality proof.

7. A GENERAL DIAMETER-THREE CRITERION

Theorem 7.1. *Let G be a connected k -regular graph of girth at least five and diameter three. If its adjacency eigenvalues are*

$$k = \theta_0 > \theta_1 \geq \cdots \geq \theta_{n-1},$$

then

$$D = 3J + (k - 3)I - 2A - A^2,$$

the principal distance eigenvalue is $3n - k^2 - k - 3$, and every nonprincipal adjacency eigenvalue θ gives the distance eigenvalue

$$\mu(\theta) = k - 2 - (\theta + 1)^2.$$

Consequently

$$G \text{ refutes WOW-284} \iff \max_{\theta \neq k} |\theta + 1| < \sqrt{2k - 2}.$$

Proof. The absence of triangles and 4-cycles means that two vertices at distance two have exactly one common neighbor. Therefore the distance-two matrix is

$$A_2 = A^2 - kI.$$

Since the diameter is three,

$$A_3 = J - I - A - A_2.$$

Now $D = A + 2A_2 + 3A_3$, which simplifies to the asserted polynomial. On $\mathbf{1}^\perp$, J vanishes, giving $\mu(\theta)$. Since regularity gives $\delta^* = k$ and the principal distance eigenvalue is positive, the strict WOW inequality is equivalent to

$$k > \max_{\theta \neq k} ((\theta + 1)^2 - k + 2),$$

which is the displayed criterion. $\square$

Corollary 7.2 (Bipartite obstruction). *A connected k -regular bipartite graph of diameter three and girth at least five is not a strict counterexample for $k \geq 3$.*

Proof. The adjacency eigenvalue $-k$ is nonprincipal. It gives

$$\mu(-k) = k - 2 - (k - 1)^2 = -k^2 + 3k - 3.$$

For $k \geq 3$, the corresponding value of $-\mu(-k)$ is at least k , with equality only at $k = 3$. $\square$

As exact negative tests, the Odd graph $O_4 = KG(7, 3)$ has

$$\text{Spec}(D) = \{82^{(1)}, 2^{(14)}, (-2)^{(6)}, (-7)^{(14)}\},$$

so $\delta^* = 4 < 7$. The Heawood graph has

$$\chi_D(x) = (x - 27)(x + 3)(x^2 + 4x - 4)^6,$$

so its least distance eigenvalue is $-2 - 2\sqrt{2} < -3$. Neither is a counterexample.

8. MOORE SECOND SUBCONSTITUENTS

Theorem 8.1. *Let M be a degree- K Moore graph of diameter two, with $K \geq 3$, and fix a vertex v . Let X be the subgraph induced by the vertices at distance two from v . Then*

$$|V(X)| = K(K - 1), \quad X \text{ is } (K - 1)\text{-regular},$$

X has girth at least five and diameter three, and

$$\lambda_{\min}(D(X)) = -\frac{5 + \sqrt{4K - 3}}{2}.$$

Consequently

$$\Phi(X) = K - 1 - \frac{5 + \sqrt{4K - 3}}{2},$$

which is positive exactly for integer $K \geq 6$.

Proof. Relative to $\{v\} \sqcup N(v) \sqcup V(X)$, write

$$A(M) = \begin{pmatrix} 0 & \mathbf{1}^\top & 0 \\ \mathbf{1} & 0 & C \\ 0 & C^\top & B \end{pmatrix}.$$

Each vertex of X has one neighbor in $N(v)$ and $K - 1$ in X , so B is $(K - 1)$ -regular. Distinct rows of C have disjoint support and row sum $K - 1$, hence $CC^\top = (K - 1)I$. The middle-right block of the Moore identity

$$A(M)^2 = (K - 1)I - A(M) + J$$

gives

$$CB = J - C.$$

Thus B has eigenvalue -1 on the $(K - 1)$ -dimensional image of $C^\top$ orthogonal to $\mathbf{1}$, while the principal eigenvalue is $K - 1$. On $\ker C$, the bottom-right block reduces to

$$B^2 + B - (K - 1)I = 0.$$

The two remaining adjacency roots are

$$r = \frac{-1 + \sqrt{4K - 3}}{2}, \quad s = \frac{-1 - \sqrt{4K - 3}}{2}.$$

The graph has diameter three; equivalently, its intersection array is $\{K - 1, K - 2, 1; 1, 1, K - 1\}$. Applying Theorem 7.1 with $k = K - 1$, the root r gives

$$(K - 3) - (r + 1)^2 = -\frac{5 + \sqrt{4K - 3}}{2},$$

which is the least distance eigenvalue. The final inequality is elementary: for integer $K \geq 3$, it becomes strict precisely at $K \geq 6$. $\square$

For $K = 7$, Theorem 8.1 is the explicit 42-vertex graph. A degree-57 Moore graph, if it exists, would produce a 56-regular example on 3192 vertices. Its existence remains unresolved; recent work still treats it as the open case [8]. Therefore this theorem is a parameterized construction, not an unconditional infinite family.

9. AN EQUITABLE-DELETION FRAMEWORK

The 40-vertex construction is part of a more general block mechanism.

Proposition 9.1. *Let M be a degree- K Moore graph of diameter two. Partition its vertices as $V(M) = R \sqcup S$ and write*

$$A(M) = \begin{pmatrix} B & C \\ C^\top & P \end{pmatrix}.$$

Assume that $M[S]$ is r -regular and every vertex of R has exactly $t > 0$ neighbors in S . Then $M[R]$ is $(K - t)$ -regular. If $Py = \eta y$ and $y \perp \mathbf{1}$, then

$$BCy = -(1 + \eta)Cy, \quad \|Cy\|^2 = (K - 1 - \eta - \eta^2)\|y\|^2.$$

Moreover, on $\ker(C^\top)$,

$$B^2 + B - (K - 1)I = 0.$$

Whenever $M[R]$ has diameter three, these adjacency data can be inserted directly into Theorem 7.1.

Proof. The upper-right and lower-right blocks of the Moore identity are

$$BC + CP = -C + J, \quad C^\top C + P^2 = (K - 1)I - P + J.$$

Apply these identities to an eigenvector orthogonal to $\mathbf{1}$. Finally, $C\mathbf{1} = t\mathbf{1}$ implies that $\ker(C^\top) \subseteq \mathbf{1}^\perp$, and the upper-left block gives the last equation. $\square$

For the 40-vertex graph, $K = 7$, S is the induced Petersen graph, $r = 3$, and $t = 1$. A counting specialization is also useful: if S itself is a degree- r Moore graph and every outside vertex has exactly one neighbor in S , then cross-edge counting forces

$$K = r^2 - r + 1.$$

The known chain $C_5 \subset \text{Petersen} \subset \text{Hoffman-Singleton}$ realizes $(r, K) = (2, 3)$ and $(3, 7)$, but it does not currently yield an infinite chain.

10. OTHER GENERALIZATION CHECKS

10.1. Balanced coordinate-layer deletions. For $m = 0, 1, 2, 3, 4$, delete the full P -layers and Q -layers with first coordinate in $\{0, \dots, m - 1\}$. Exact representative calculations give:

m	$ V $	degree	$\lambda_{\min}(D)$	Φ
0	50	7	-4	3
1	40	6	-5	1
2	30	5	-6	-1
3	20	4	-7	-3
4	10	3	-3	0

Thus the natural layer-deletion chain produces the 50- and 40-vertex strict counterexamples, but does not continue to smaller strict examples.

10.2. **What remains open.** The checks in this package cover:

- (i) all regular diameter-two girth-five graphs through the Moore mechanism;
- (ii) every regular girth-at-least-five diameter-three graph through the exact adjacency-spectrum criterion;
- (iii) Moore second subconstituents;
- (iv) equitable induced-subgraph deletions, including the Petersen deletion;
- (v) all one-vertex and edge-endpoint deletions of the fixed 40-vertex graph;
- (vi) the balanced layer-deletion chain, the Odd graph O_4 , and the Heawood graph.

No unconditional infinite family is established. In particular, fixed-degree diameter-three graphs have bounded order by the Moore bound, so an infinite family through Theorem 7.1 would require unbounded degree or a new operation outside the diameter-three setting. Covers, subdivisions, and vertex-replacement operations are not asserted to preserve the strict inequality; no exact preservation theorem was found for them.

11. REPRODUCIBILITY AND CERTIFICATE SEPARATION

Run, from the package root,

```
python scripts/verify_wow284_38_40_42.py
python scripts/verify_38_graph6_independent.py
python scripts/verify_descendant_families.py
python scripts/explore_generalizations.py
```

The first script reconstructs the ambient coordinate graph and verifies the 42-, 40-, 39-, and 38-vertex results. The second starts only from graph6, uses NetworkX shortest-path routines, and implements a separate exact **Fraction**-based LDL^T algorithm. The third exhausts the 40 singleton and 120 edge deletions with exact characteristic polynomials and Sturm counts. The fourth contains exact family checks and one clearly marked floating-point screen. SymPy and NetworkX are cited in [9, 10].

Machine-readable graph6 strings, complete adjacency lists and edge lists are in `data/graphs/`. The package includes captured outputs and SHA-256 checksums. These are reproducibility certificates, not external peer review or a claim of priority.

APPENDIX A. EXACT DISTANCE CHARACTERISTIC POLYNOMIALS

For the 39-vertex graph,

$$\begin{aligned} P_{39}(x) = & x^9(x+5)^{12}(x^2+6x+3) \\ & \cdot (x^3+3x^2-15x-7)^2(x^3+3x^2-15x-3)^2 \\ & \cdot (x^4-78x^3+303x^2-70x-450). \end{aligned}$$

For the 38-vertex graph,

$$\begin{aligned} P_{38}(x) = & x^4(x-2)(x+5)^8(x^2+6x+2)^2 \\ & \cdot (x^3+3x^2-15x-3) \\ & \cdot (x^4+5x^3-7x^2-23x-6) \\ & \cdot (x^5+9x^4+7x^3-77x^2-54x-4) \\ & \cdot (x^9-67x^8-404x^7+1772x^6+7205x^5 \\ & \quad - 18489x^4-17018x^3+20288x^2+16824x+1680). \end{aligned}$$

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ÉCOLE NORMALE SUPÉRIEURE, UNIVERSITÉ PSL, PARIS, FRANCE
Email address: `samuil.petkov@ens.psl.eu`